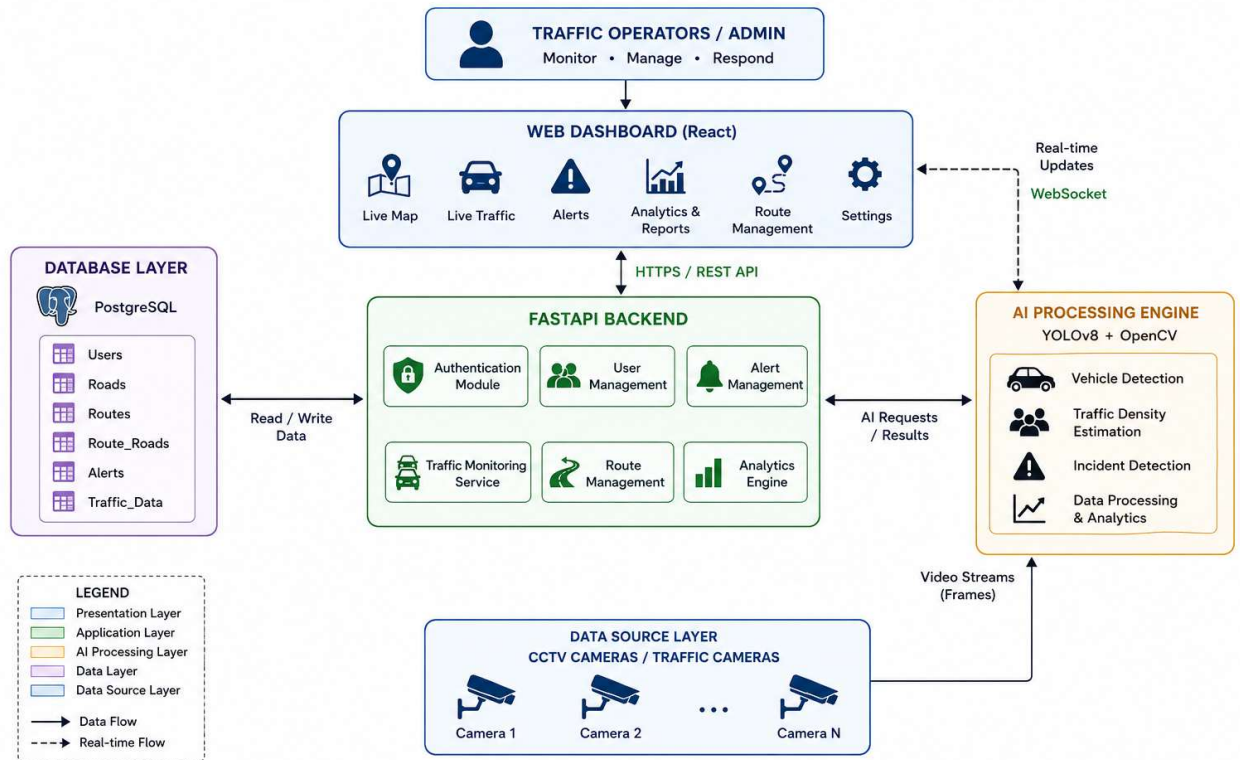


TRAFFICVISION AI – SYSTEM ARCHITECTURE



Architecture Overview

This architecture illustrates the overall workflow of the TrafficVision AI system. Traffic cameras capture live video streams, which are processed by an AI engine (YOLOv8 and OpenCV) to detect vehicles, estimate traffic density, and identify incidents. The processed results are sent to the FastAPI backend, where they are stored in a PostgreSQL database and made available to the React web dashboard through REST APIs and WebSockets.

Components

Component	Description
Traffic Operators/Admin	Monitor traffic, manage routes, and respond to incidents.
React Dashboard	Displays live traffic, alerts, analytics, reports, and settings.
FastAPI Backend	Handles authentication, APIs, business logic, alerts, routes, and analytics.
AI Processing Engine	Uses YOLOv8 and OpenCV for vehicle detection, density estimation, and incident detection.
PostgreSQL Database	Stores users, roads, routes, alerts, and traffic data.
CCTV Cameras	Provide live video streams for AI processing.